Predicting SNAP Payment Errors: A Pilot Proposal for Utah

Synthetic-data-validated case prioritization for quality-control review — the low-error panel's decisive test

October 4, 2026 · Research whitepaper · Methods and validation artifacts are open-source at github.com/ctrimm/snap-qc-per-experiments

1. Executive summary

Utah has the third-lowest SNAP payment error rate in the nation (5.54% FY2025, federally published) — the decisive third leg of our low-error panel alongside South Dakota (lowest) and Idaho (second-lowest). We built and validated a Utah-conditioned synthetic data generator, calibrated against federal quality-control (QC) microdata (4,101 Utah cases, FY2020–2024; fitted on FY2022–2024, excluding partial-year files). Machine-learning models trained *only* on this synthetic data rank real Utah cases by error risk well above chance — finding AUC (a ranking metric where 0.5 is chance and 1.0 is perfect) of **0.728 vs 0.678** for a nationally trained model — and the advantage survived a fully clean refit that excluded every holdout row. Proof the synthetic cases carry genuine signal about where errors happen, even in one of the nation's cleanest caseloads.

The operational finding — and the panel's decisive result: the winning ranking rule is the *two-part* P(error) × expected-dollars model, recovering **$4,604 vs $2,701 for random review** at a fixed review budget — **$45 vs $26 per hour of reviewer time**. This does *not* replicate South Dakota's surprise, where probability-alone won: Utah's dollars, small on average, concentrate in a heavy tail (the top fifth of cases holds 93% of all error dollars), and the dollar-aware model finds that tail. Reviewing the riskiest half of cases catches 74% of all errors with half the reviews.

We propose a low-risk pilot: train the ranker on Utah's own historical QC data, validate on a recent holdout, then run it in shadow alongside existing review. No production eligibility-system changes. No new data collection.

2. The problem

SNAP payment error rates remain stubbornly high nationwide, and every error is either taxpayer money over-issued or a household underpaid. Quality-control review is the corrective mechanism — but reviewer time is finite, and cases are currently selected for review without a data-driven sense of where errors concentrate. The result is expensive review effort spread thinly over a caseload where risk is highly uneven.

Utah is the low-error panel's decisive test. Our analysis of 4,101 Utah QC cases (FY2020–2024) shows the state's benefit-calculation engine matches federal statute at 95.6–99.2% exact per fiscal year — Utah's errors live in the *human process layer* (verification, timing, documentation), which is precisely where a prediction tool adds value. The error signature is distinctive: wage-timing errors — earnings the agency was not yet required to act on — run at 1.95x the national rate and make up a fifth of all error weight; shelter-deduction errors cut both directions (over-deductions at 2.21x national, under-deductions at 1.33x); substantive errors are coded to the client at 1.56x the national rate, with essentially no "unknown cause" coding. Error dollars are small (median $22 vs $31 nationally, 90th percentile $99 vs $202) but concentrated: the top tenth of errors by dollar size holds 75% of all error dollars, earned-income and shelter cases dominating the tail.

3. What was built and validated

A Utah-conditioned synthetic generator. Starting from a nationally calibrated generator, we fit Utah-specific error-rate offsets and empirical error mixes on FY2022–2024 microdata (partial-year FY2020–21 excluded — a lesson from the New York track). The UT-conditioned output reproduces Utah's finding mix (L1 0.045), element mix (L1 0.070), and cause mix (L1 0.035 — the tightest of the four states studied), while leaving national output byte-identical. Documented residuals: the generator slightly overstates the over-issuance share (+6.5pp), a structural limit of its direction machinery that downstream calibration absorbs.

Validated predictive signal. Models trained only on UT-conditioned synthetic cases predict real, held-out Utah errors better than the nationally trained model on all three tested seeds: finding AUC 0.724 vs 0.649 on average (the largest finding edge of any state studied), and the edge survived a fully clean refit excluding every holdout row (pristine AUC 0.728 vs 0.678 — no contamination premium). On fine-grained error-type classification the Utah model wins the majority of comparisons but the margin is thin and seed-dependent; the transferable signal lives in the finding and dollar tasks, and the whitepaper's claims are scoped accordingly.

Two-part wins — the South Dakota reversal does not replicate. In South Dakota, ranking by error probability alone won (2.07x) because errors were uniformly tiny and dollar prediction added only noise. In Utah, probability-alone manages just 1.08x — the *worst* of the three rules — while the

two-part model reaches 1.70x. The mechanism is visible: Utah's dollars concentrate in a heavy tail (93% in the top fifth of cases) that the expected-dollar leg successfully targets. The panel lesson is refined: it is not "low error rate implies probability-alone wins" — it is "no dollar tail implies probability-alone wins." The ranking rule must be re-fit to each state's error economics, not copied between states.

> **Headline pilot economics (simulated on held-out Utah cases, top-20% review budget):**
>
> | | Random review | Two-part ranker |
> |---|---|---|
> | Error dollars recovered | $2,701 | **$4,604** |
> | Value per review hour | $26 | **$45** |
> | Share of all error dollars captured | — | 34.1% |
>
> Of the $4,604: $3,124 is over-issuance (direct program savings); $1,480 is under-issuance (households underpaid — the correct QC outcome, but a make-whole cost). Savings-only value: $30/hr vs ~$15/hr for random review. The synthetic-trained ranker captures 89% of the performance a model trained on real Utah data is projected to reach.

4. The pilot

What Utah provides

- Historical QC case extracts the state already holds (no new data collection, no new PII).
- A review team willing to work a ranked queue for a defined pilot period.

What changes — and what does not

- **Phase 0 (cold start, no Utah data needed):** the synthetic-trained ranker scores a sample of cases; rankings are validated against known outcomes.
- **Phase 1:** the ranker is trained on Utah's historical QC data and validated on a recent holdout the model never saw. Phase 1 also selects the ranking rule for Utah's economics (two-part vs

probability-alone vs direct) rather than inheriting another state's, and runs the household-type review-burden analysis as a deployment gate (see limitations).

- **Phase 2 (shadow):** the ranker scores the live review caseload; reviewers work the top slice; outcomes are measured against the baseline selection method.
- **What does not change:** no modifications to the eligibility system, no automated determinations (the model prioritizes review — it never decides a case), no data leaves the state's control.

Success metrics

- Precision in the reviewed slice vs baseline selection.
- Error dollars recovered per reviewer hour vs baseline.
- Reviewer time saved to reach a fixed error-capture target.

A reasonable pilot shape is 8–12 weeks from data handoff to shadow-readout.

5. Limitations and honest caveats

- **Small samples, wide intervals.** The holdout contains 337 error cases; per-seed margins are thin and confidence intervals are wide. We report three seeds and prefer qualified claims — the finding-task transfer is consistent across seeds, but this is a small-sample result and should be re-validated on Utah's own larger historical extract in Phase 1.
- **The ranking rule is state-specific.** Two-part wins in Utah because a heavy dollar tail exists for the expected-dollar leg to find; in South Dakota, probability-alone won because there was no tail. If Utah's error economics shift, the rule must be re-tuned — Phase 1 selects the rule per state rather than assuming transfer.
- **Refit on a rolling multi-year window; no annual refit required.** The transfer signal is year-to-year stable (no decay across the FY2022→FY2023–24 boundary), but two shifts deserve watches: the medical-deduction bucket showed real F1 decay, and FY2024 brought a coding-practice change (Utah began recording causes on timing errors — the blank rate dropped from ~0.98 to 0.61). Partial-year files are excluded from refits.
- **Procedural errors are invisible.** The federally published payment error rate (5.54% FY2025) and the case-microdata-implied case-error rate (33.2%) measure different things: the published rate is dollar-weighted and includes procedural errors the microdata cannot see. This tool targets substantive case errors, not procedural compliance.

- **Headroom.** The synthetic-trained two-part ranker reaches 1.70x vs 1.92x for the real-trained projection — the generator's feature-to-dollar conditional is the binding constraint, and the pilot's Phase 1 exists to close exactly this gap.
- **Human judgment stays in the loop.** This is a triage tool for reviewers, not an adjudication engine. Final determinations remain human — by design.

Review-burden note

Unlike South Dakota — whose winning rule was probability-alone — Utah's recommended rule is dollar-maximizing, so the risk-regressivity caveat from the Pennsylvania and New York tracks applies in full: at a 20% review budget, elderly and disabled households (the highest-error groups, at 36–41% base error rates) are reviewed at roughly one-sixth the rate their error risk warrants (review-to-risk ratios of 0.155), while the lowest-risk childless-adult households are over-reviewed at 1.8x. The expected-dollar leg drives this tilt (probability-alone is more than twice as proportional). A household-type-stratified review-burden analysis is a required Phase-1 deployment gate, with per-type review floors as the operational mitigation — the dollars recovered must not come at the cost of abandoning review where errors are most frequent.

6. The ask

We are looking for a Utah partner to run this pilot: a data contact who can provide historical QC extracts, and a review team for the shadow phase. We bring the validated methodology, the synthetic cold-start model, and the open-source validation artifacts. All methods, experiments, and results are published openly; nothing about the approach is proprietary or black-box.

Methods, experiment chronicles (70+), and validation code: github.com/ctrimm/snap-qc-per-experiments (private research repo; access on request). This work is conducted in the open-source civic-technology lane. Contact: via the repository.